Calculation of Bosten Lake Water Storage Based on Multiple Source Remote Sensing Imagery

Weiwei Wang, Fei Zhang, Jingchao Shi, Qi Zhao, Changjiang Liu, Mou Leong Tan, Hsiang-Te Kung, Guang Gao, and Gang Li

***Abstract*— Bosten Lake is a crucial water source in arid northwest China, which has been maintaining the ecological balance of the southern Xinjiang region. It contributes to the sustainable development of the local economy and watershed ecology. Insufficient hydrological data for the basin, however, cause uncertainty in hydrological modeling and makes it difficult to calculate water availability and lake-water storage using conventional methods. Therefore, this article proposed a novel method to retrieve the lake's water depth from Landsat-8 OLI and ICESat-2 satellite data using neural network (NN) model. Specifically, the Rayleigh-corrected top-of-atmosphere (TOA) reflectance (ρ_{rc}) in the 443–2300 nm range was employed as the input to the NN model for the retrieval of water depth. This avoids the requirement to correct the effects of aerosols. In addition, the water depth retrieved by the NN model was compared with a conventional dual-band ratio model (DBRM). To evaluate the accuracy of the corrected ICESat-2 photon data, the in situ water depths were compared with the corrected ICESat-2 photon data (considering water-level variations at different times), and 20 in situ water depth values near the ICESat-2 track were selected for verification. The results showed that 1) the in situ water depth and the corrected ICESat-2 photon data had the coefficient of determination (R^2) values of 0.94; 2) NN inversion of bathymetry presented a feasible method of obtaining regional bathymetry, with R^2 of 0.87, root mean square error (RMSE) being 1.17 m, and mean absolute error (MAE) being 0.26 m. In contrast, DBRM generated the R^2, RMSE, and MAE values of 0.74, 1.62, and 0.93 m, respectively; and 3) the water storage in Bosten Lake's from April to September in 2022 ranged from 6.73×10^9 to 7.50×10^9 m^3. The findings can provide a strong scientific basis for the sustainable development and the rational allocation regional water resources.**

***Index Terms*— Arid region, Bosten Lake, multiple remote sensing, neural network (NN) model, ungauged watershed, water depth and storage.**

I. Introduction

LAKES are important parts of the hydrological cycle. Lake characteristics and changes can affect regional ecology, biodiversity, aquatic organism conservation, economic development, and human well-being in the local area [1]. Lake water storage variables are important indicators or sentinels of the environmental impacts of climate change and human activities [2]. They are essential for accurately assessing changes in watershed climate and hydrological environments [3]. Lakes are important components of surface water bodies, playing crucial roles in human life and production. Lakes are rich freshwater resources, which furnish important support for regional development [4]. However, climate change and human activities have been disturbing the spatial and temporal distribution of water resources. Such impacts induce frequent floods and droughts and amplify lake water storage fluctuations, harming regional economic and social welfare and lake-ecosystem health [5]. Freshwater lakes in arid regions perform special functions as valuable and limited freshwater resources and maintain regional ecological and environmental quality [6], [7].

Recent advances in remote sensing technology have promoted the increasing use of remote sensing data to characterize the storage variables of lakes quantitatively [8]. This method enlists digital elevation models or remote sensing data and satellite altimetry data to extract lake area and water level, construct a relationship curve between them, and combine mathematical geometric models to estimate the lake's storage volume [9]. The broad coverage of remote sensing data with little geographical limitation permits large-scale and rapid surveys of lake storage variables [10], [11]. However, limitations in quality, quantity, and time span of remote sensing data suppress the extraction accuracy of water level and lake area as well as the estimation of lake storage variables [12]. In addition, due to the sparse orbits of altimetric satellites, necessary observational data are often insufficient for small

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and medium-sized lakes. The recent launch of many remote sensing satellites has improved the coverage of lakes, offering chances for comprehensive and efficient investigations of lake storage variables over long time series [13].

Water depth assessment is a crucial aspect of lake-water storage studies [14], [15]. Traditionally, bottom depth measurements are made by single-beam bathymetry and multibeam echo sounders [16], [17]. Single-beam bathymetry is low-cost and simple to operate, but it can acquire only one bathymetry point at a time, which is inefficient and cannot provide a full-coverage measurement. Multibeam bathymetric systems can acquire hundreds of bathymetric points in one measurement, achieving full-coverage and high-accuracy measurement of underwater terrain [18]. These methods can provide highly accurate bathymetric data [19], [20]. However, they are costly, time-consuming, and limited to areas accessible by ships or aircrafts. The resulting data from space are difficult to transform into high-resolution bathymetric maps. In addition, the application of an airborne LiDAR bathymetry system is restricted by its system price. Thus, its market share remains low [19].

The recently launched ICESat-2 has an advanced laser altimeter system, which provides a new method for bathymetry [21]. Meanwhile, many satellites with good spatial resolution imagers routinely measure lake depths. By using empirical regression, it is feasible to generate bottom depth maps from ICESat-2 and imager data [21]. The dual-band ratio model (DBRM) is a typical example that uses the ratio of the remotely sensed reflectance (R_{rs}) of two bands to estimate water depth and derives an empirical coefficient of DBRM for each image [22]. The DBRM is widely used in optical shallow water bathymetry, both onshore and offshore [23], [24]. However, the use of two spectral bands generates nonuniqueness problems, leading to large uncertainties, especially when the nature of the bottom substrate or water changes significantly within a scene [25].

With the development of the computer technology, neural network (NN) offers a powerful method for accurate water depth retrieval. Unlike traditional band ratio model, NN is flexible in various inputs [26]. It can detect potentially complex nonlinear relationships without prior knowledge and has been widely used to monitor water depths in marine, coastal, and inland waters [27]. However, for inland Class II water bodies, the NN model is sensitive to atmospheric correction algorithms [28]. With no valid remote sensing reflectance (R_{rs}) due to the failure of atmospheric correction, no bathymetric data are available for these locations. Alternatively, this study used Landsat-8 data, ICESat-2 data, and a NN to retrieve the Bosten Lake water storage. To avoid the failure of atmospheric correction, Rayleigh-corrected top-of-atmosphere (TOA) reflectance (ρ_{rc}) was used as the model input to retrieve the lake's water depth.

The Bosten Lake, our research object, is situated in an area typically lacking hydrological observation stations. The watershed is an important base for commercial food production [29]. The local economy mainly depends on the primary activities of livestock rearing and crop cultivation. Urban domestic water and agricultural irrigation rely heavily on river and lake water. However, traditional lake storage calculations are time-consuming, labor-intensive, and limited to places that can be reached by vessels or airplanes. In particular, the traditional algorithms for bottom depth are hardly portable to new measurements. Therefore, we present a novel method to retrieve the optical shallow lake bathymetry directly from ICESat-2 and Landsat-8 OLI, combined with the NN model. More importantly, we used Rayleigh-corrected TOA reflectance (ρ_{rc}) data as input to the NN model for retrieval of water depth to avoid atmosphere correction failure. In addition, the water depth retrieved by the NN model was compared with a conventional DBRM. Then, the optimal water depth retrieval model NN was used to calculate Bosten Lake's water storage to provide a basis for water-resource control and management in the small basin. The specific research objectives are 1) to assess the potential of ICESat-2 in-depth monitoring of inland optical shallow lakes and 2) to analyze the accuracy of water depth calculation by a NN model.

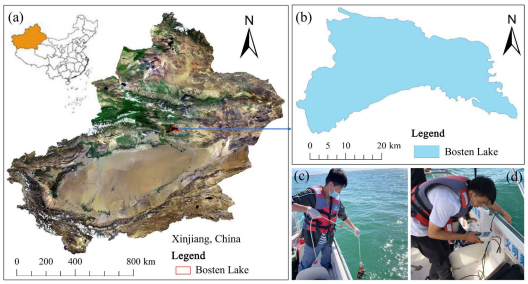


Fig. 1. (a) Xinjiang Uyghur Autonomous Region in northwestern China. (b) Bosten Lake. The bathymetry extraction using (c) sounding hammer and (d) depth sounder.

II. Materials and Methods

A. Study Area

Bosten Lake is located at the southern foot of the Tianshan Mountains in Bohu County, Xinjiang, northwest China (Fig. 1). Its annual average temperatures range from 8.2 °C to 11.5 °C, with the lowest in January, the highest in July, and the maximum extreme exceeding 40.0 °C [30], [31]. The annual precipitation is about 70 mm, mainly dropped from May to September, and the annual pan evaporation is over 1880 mm. The lake marks the confluence of large and small rivers in the Bosten Lake watershed, with over 90% of the water supplied by the Kaidu River and the rest by smaller rivers such as Huangshui, Qingshui, Quhui, and Ushtara Rivers [29]. The study area is characterized by a varied and complex terrain, including mountains, basins, desert Gobi, river valleys, and plains. Bosten Lake is a Mesozoic depression lake with a low flat middle basin [29]. In paleoclimate and environmental evolution, Bosten Lake is an important indicator of environmental evolution in arid Central Asia [32]. Global changes and human activities brought deleterious ecological impacts on the lake and the watershed, imposing water-area shrinkage, water pollution, and soil salinization.

On June 27, 2022, the water depth was measured in Bosten Lake. The in situ sampling was carried out with a sounding

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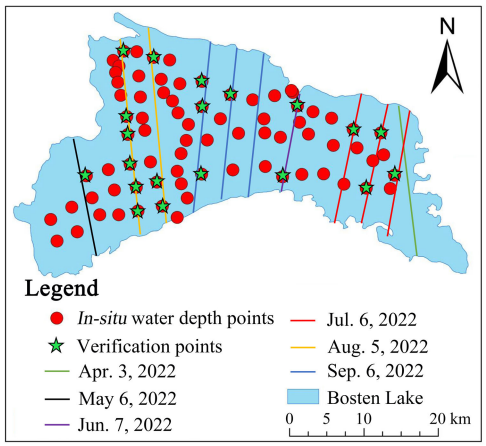


Fig. 2. In situ water depth points and matched ICESat-2 track.

hammer and a hand-held ultrasonic sounder (FW-SFCC) [Fig. 1(c) and (d)]. The sounding hammer was used to calibrate the hand-held ultrasonic sounder. The sampling points are shown in Fig. 2. As Bosten Lake freezes during December to February, March to November is the selected time for field measurements.

B. Data Sources

1) Satellite Image Data: Satellite image data, obtained from the U.S. Geological Surveyhttps://earthexplorer.usgs.gov/), was radiometrically and geometrically corrected by the system. The Landsat-8 OLI was processed using the SeaDAS package (v8.2.0) to obtain the Rayleigh-corrected TOA reflectance (ρ_{rc}) as calculated by the following equation:

$$\rho_{rc} = \frac{\rho_t}{t_g} - \rho_r = \frac{\pi L_t/(F_0 \cos(\theta_s))}{t_g} - \rho_r \tag{1}$$

where, ρ_t represents the TOA total reflectance, F_0 represents the extraterrestrial solar irradiance, θ_s represents the solar zenith angle, t_g represents the gas transmittance, and ρ_r represents Rayleigh scattering reflectance.

2) Satellite Altimetry Data: ICESat-2, launched on September 15, 2018, carries a photon-counting laser altimeter. It has three laser beams along the orbit with a cross-track distance of 3.3 km between adjacent beams. Each laser beam is divided into two sub-beams, i.e., a strong beam with pulse energy about four times higher than the weak beam [2]. This study used ICESat-2 ATL03 geolocation photons (https://search.earthdata.nasa.gov/). The Level-2 ATL03 dataset included all the original photons recorded in six different trajectories, i.e., three strong and three weak beams, each with information such as latitude, longitude, and altitude [33].

With the assumption that the water depth is constant over a short period, the ±7 days period between Landsat-8 and ICESat-2 were used to match (Table I). To match measurements between Landsat-8 and ICESat-2, the ICESat-2 track were located in the Landsat-8 image. For the ICESat-2 corrected bottom depth points, calculations were made with reference to literature [9].

TABLE I
MATCH DATES FOR LANDSAT-8 AND ICESAT-2

Data sources	Period
Landsat-8	April 7, 2022., May 1, 2022., June 2, 2022., July 13, 2022., August 5, 2022., September 6, 2022
ICESat-2	April 3, 2022., May 6, 2022, June 7, 2022., July 6, 2022., August 5, 2022, September 6, 2022

C. Methods

1) Lake Area Extraction: Landsat-8 OLI images with less than 10% cloudiness were selected, and radiometric calibration and atmospheric correction were performed. Modified Normalized Difference Water Index (MNDWI) has been proven to be the most effective method of differentiating water body with other land features, with advantages in resisting mountain shadows [12], [32]. This method has been widely employed in surface water body studies because of its effectiveness in reducing underly noise [34], [35]. This study extracted the water surface information by the MNDWI equation, as expressed in the following equation:

$$\text{MNDWI} = (\text{band}_{\text{Green}} - \text{band}_{\text{MIR}})/(\text{band}_{\text{Green}} + \text{band}_{\text{MIR}}) \tag{2}$$

where $\text{band}_{\text{Green}}$ is the reflectance of the green band, and band_{MIR} is the reflectance of the mid-infrared band.

After calculating and obtaining the MNDWI images, the error caused by the artificial rate threshold was reduced by the Otsu method (OTSU) to automatically determine the water decent threshold recognition [36]. The OTSU algorithm was used to extract the watershed area using high-resolution imagery available on Google Earth (www.earth.google.com), and the nonlake main body area was eliminated [37]. Then, the extracted water body layer was imported into the ArcGIS10.4 software (ESRI, Redlands, CA) to eliminate the water data of nonlakes. Finally, the lake's surface area was calculated using the projection of the WGS 1984UTM45N area.

2) Water Depth Calculation: From Fig. 3(a), when the sensor is collecting photons, the raw ATL03 data contain a large amount of noise due to the interference of background noise, which requires point cloud denoising and signal extraction. The density-based spatial clustering of applications with noise (DBSCAN) is a typical denoising algorithm based on point cloud density that can effectively handle noisy points and discover spatial clusters of arbitrary shapes [33]. At present, the DBSCAN algorithm is mainly used in the elimination of photon point cloud noise in the level 2 standard data product ATL03 of ICESat-2 [24]. The DBSCAN method is widely used to remove outliers from ATL03 photon data [33], [38]. Therefore, we used adaptive threshold the minimal number of points (*MinPts*) to extract the signal photons on ATL03 from the adaptive DBSCAN calculator of the original data photons of ATL03. As shown in Fig. 3(b), the adaptive DBSCAN algorithm extracts signal photons from the raw ATL03 data. In the DBSCAN algorithm, points in a cluster are classified as signals when the point density of their neighboring points exceeds the *MinPts* [24]. From Fig. 3(c), the actual data tracking method was employed for the optically deep area.

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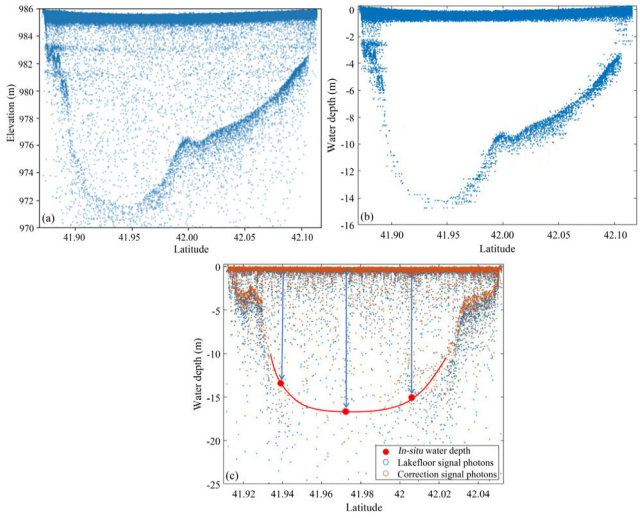


Fig. 3. Examples of (a) raw ICESat-2 photon data and photon data of (b) surface and bottom after the filtering procedure, collected on May 6, 2022; tracking method of (c) measured data ICESat-2 data collected on July 6, 2022.

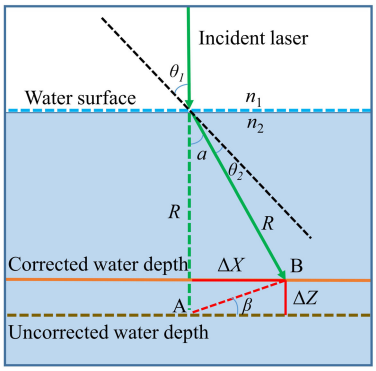


Fig. 4. Correction for the refraction bias of water depth arising from the lake surface. θ_1 is the incidence angle, θ_2 is the refraction angle in the water column, a is the angle of refraction R is the laser range in the water column, and ΔZ is the refraction bias, n_1 is air refractive index, n_2 is waterbody refractive index.

3) Water Depth Calibration: The ICESat-2 product does not consider the refractive index of the air–water interface and speed of light variations. This phenomenon produces errors in the geographic position recorded in ATL03, resulting in the position being further and deeper from the nadir than the actual situation [24]. Therefore, the ICESat-2 mapping requires a refraction correction. The incidence of the three laser beams of ICESat-2 is assumed to be close to the nadir, as shown in Fig. 4.

This study used the refraction correction method that was proposed by literature [39]. The final return underwater photon depth (H) was calculated by the following equation:

$$H = Dep - \Delta Z = Dep - \tan[(\theta_1 - \theta_2)/2] * \sin(\theta_1 - \theta_2)R \tag{3}$$

where Dep is the depth of the uncorrected underwater photons detected, R is the corrected tilt range, θ_1 is the angle of incidence, and θ_2 is the refraction angle in the water column.

4) NN Model: NN model is capable of handling nonlinear separable problems and is easy to implement, fast, scalable, and highly stable [40]. NN model is composed of an input layer, hidden layers, and an output layer. The seven bands of Landsat-8 OLI were used as the input layer. The number of hidden layers and neurons in each layer was obtained through trial and error. Fig. 5 shows our NN algorithm designed to process the water bottom depth that can be retrieved from ρ_{rc}. To train the NN, April 8, 2022, May 6, 2022, June 2, 2022, July 6, 2022, August 5, 2022, and September 6, 2022, corrected water depth (Fig. 2), and the matching ρ_{rc} were selected. Since the lake water level can change at different periods (Fig. 6), the water level extraction algorithm shown in Fig. 7 was adopted in this study [18]. The results of water level extraction are shown in Table II.

TABLE II
EXTRACTED WATER LEVEL OF THE BOSTEN LAKE

Matching period	Water level (m)
April 3, 2022	1046.42
May 6, 2022	1047.49
June 7, 2022	1047.5
July 6, 2022	1047.63
August 5, 2022	1047.32
September 6, 2022	1047.15

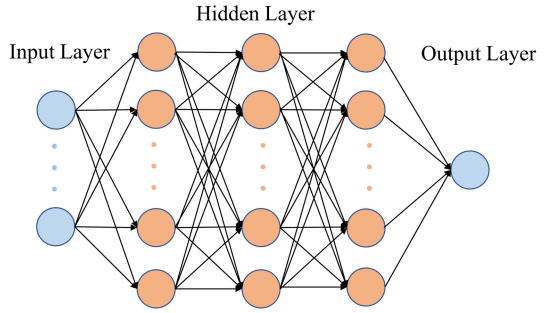


Fig. 5. Schematic chart of the study's NN algorithm.

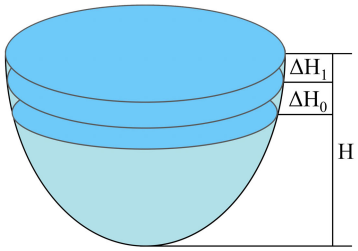


Fig. 6. Schematic of water level change in different periods.

The 3860-training dataset pairs were divided into subsets of 80/10/10 training/testing/verification to verify that whether the trained model is over-or under-fit based on the classification criteria of Lai et al [38].

5) Dual-Band Ratio Model: The DBRM was applied to obtain maps of bottom depth [23]. The ratio model is expressed as follows:

$$Z = m_1 \frac{\ln(nR_{rs}(\lambda_i))}{\ln(nR_{rs}(\lambda_j))} - m_0 \tag{4}$$

where Z is the estimated depth, n is a fixed coefficient, $R_{rs}(\lambda_i)$ and $(nR_{rs}(\lambda_j)$ are the reflectance of water surface for bands

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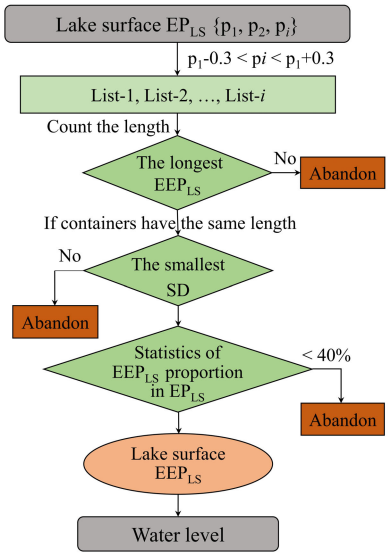


Fig. 7. Flowchart of water level extraction. SD represents standard deviation, EP_{LS} represents the elevation points of the lake surface, EEP_{LS} represents the effective elevation point.

i or j, m_1 and m_0 are tunable constants to linearly transform (slope and intercept, respectively).

D. Accuracy Assessment

The accuracy assessment was performed using the coefficient of determination (R^2), the root mean square error (RMSE) and mean absolute error (MAE) were chosen as the accuracy evaluation methods. The calculation formulas of RMSE and MAE are shown in the following equations:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(H_i - H_j)^2} \tag{5}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{H_i - H_j}{H_j}\right| \tag{6}$$

where, H_i is the bathymetry of ICESat-2 at the ith monitoring point; H_j is the inverse bathymetry value of the jth monitoring point; and n is the number of selected monitoring points.

E. Water Volume Calculation

Given the surface parameters of water area and water depth data, the lake's water volume can be estimated based on a priori knowledge of the underwater topography (Fig. 8) using the following equation:

$$V = \int_0^A H(A)dA = \int_0^A A(H)dH \tag{7}$$

where V is the storage volume (in m^3), A is the surface water area (in m^2), and H is the depth (in m).

III. Study's Conceptual Framework

Fig. 9 shows the study's conceptual framework. The first step includes collection of remote sensing and radar data and preprocessing. Next, the lake area was extracted, and ICESat-2 photons were denoised and corrected. The water depth was

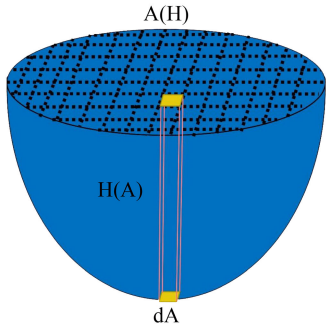


Fig. 8. Diagram illustrating the calculation method for the lake's water storage volume.

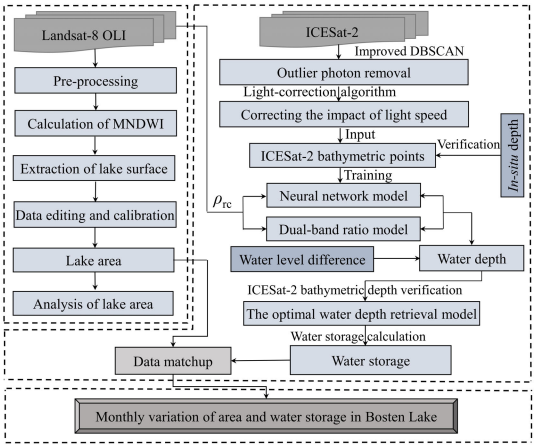


Fig. 9. Study's conceptual framework and workflow.

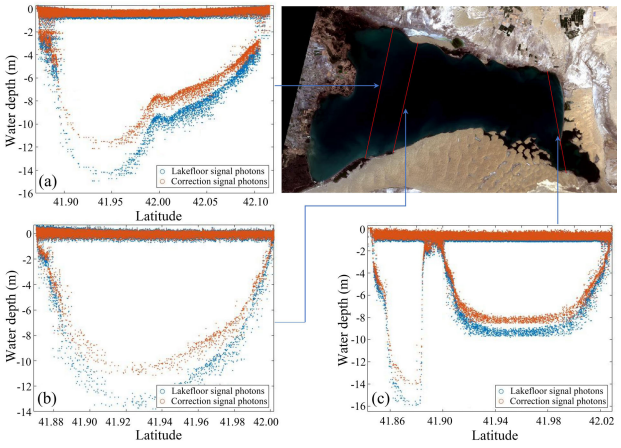


Fig. 10. Examples of the corrected underwater photons in (a) and (b) collected on May 6, 2022, and (c) collected on April 3, 2022.

calculated using the optimal water depth retrieval model and extended to calculate water volume. Finally, the water storage capacity of Bosten Lake was calculated.

IV. Results and Analysis

A. Bathymetric Retrieval

The ICESat-2 signal photons at the lake bottom were retrieved based on the adaptive DBSCAN algorithm. Fig. 10 shows the ICESat-2 lake trajectory image and the corresponding detected lake water depth with and without refraction

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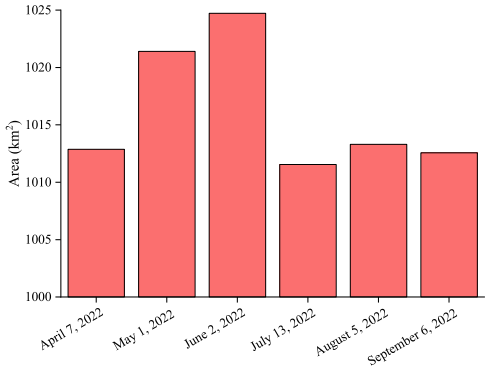


Fig. 11. Variation of Bosten Lake area.

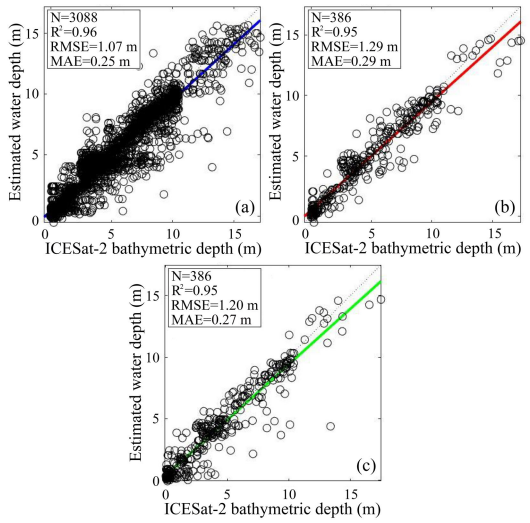


Fig. 12. Comparison between ICESat-2 bathymetric depth and NN model estimated depth of (a) training subset and (b) test subset, and validation subset. The dotted line denotes the 1:1 line.

correction. The ICESat-2 detects significant lake bottom signal photons based on the underwater topography. For each laser beam, the refraction effect in the water column was corrected using (3). The maximum depth of photons returned to the bottom of the lake after refraction correction was about ∼14 m., while the ICESat-2 retrieved signal photons at the bottom of the lake were close to ∼16 m. Therefore, without the refraction correction for vertical height, it will lead to errors in the bathymetry results. Fig. 10(c) shows that the signal photons detected by ICESat-2 at the bottom of the lake are close to ∼16 m, with clear lake water due to low human interference.

B. Variation in the Bosten Lake Area

After using MNDWI for water body identification, the OTSU was used to determine the water surface threshold recognition automatically, and the area of Bosten Lake in different seasons in 2022 was obtained (Fig. 11). The maximum lake area of 1024.73 km^2 was attained in June of 2022, while the minimum of 1011.55 km^2 was attained in July of 2022.

C. Bottom Depth of Bosten Lake

1) NN Model Retrieval Bottom Depth: Fig. 12(a) shows the ICESat-2 bathymetric depth, and NN model that estimated

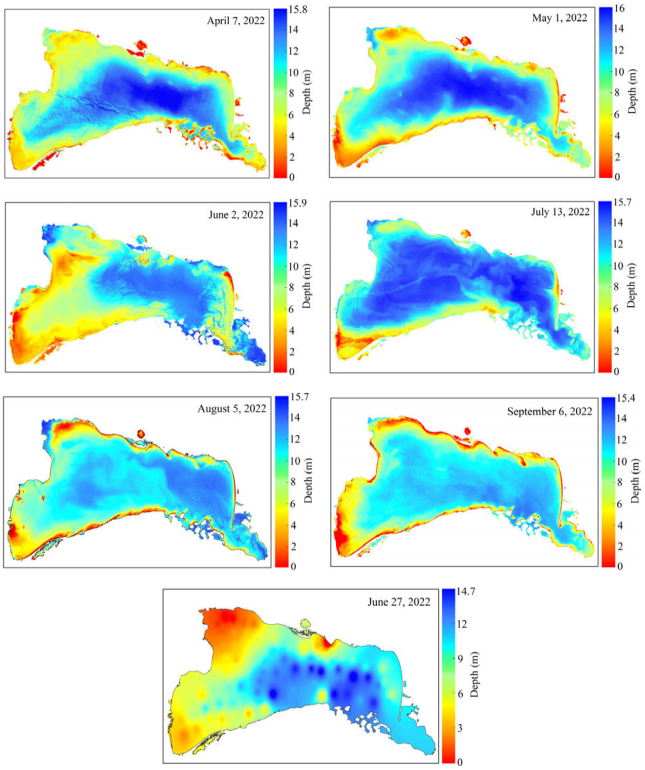


Fig. 13. Examples of the retrieved lake water depth by using NN model. The IDW method was applied to interpolate water depth on June 27, 2022.

lake depth for the subset of 3281 data was points trained by the NN. For these multitemporal images with different water properties, aerosol, and bottom features, the NN had an R^2 of 0.96 between 0 and 16 m, RMSE of 1.03 m, and MAE of 0.25 m, indicating that the NN predictions are reliable. In addition, a test subset (579 data points) was used to evaluate the NN, and ICESat-2 basic depth and NN model that estimated depth were compared, as shown in Fig. 12(b). With the training subset, most data points were situated near the 1:1 line, with R^2 of 0.95, RMSE of 1.12 m, and MAE of 0.29 m in the 0–17 m range. ICESat-2 basic depth and the estimated depth from the NN model were compared, as shown in Fig. 12(c), using a validate subset of 386 data points. Most of the data points were situated near the 1:1 line, with R^2 of 0.95, RMSE of 1.20 m, and MAE of 0.27 m in the 0–17 m range.

The water depth of the whole Bosten Lake was inverted separately using the trained NN model, and the bottom depth results are shown in Fig. 13. The water depth was between 0 and 16 m. The model generated better estimation in the lake's middle part but was less effective at the land–water interface shore. The water-depth overestimation was especially evident in the lake's northwestern part, which may be due to the low number of matches and some outliers that were caused by the matching problem, compromising the accuracy of the statistical evaluation [41]. In addition, the lake's northwestern part has quite complex spatial variation at the land–water intersection, making it difficult to match data. Thus, its spatial heterogeneity would reduce the accuracy of statistical evaluation. To further compare the efficacy of the

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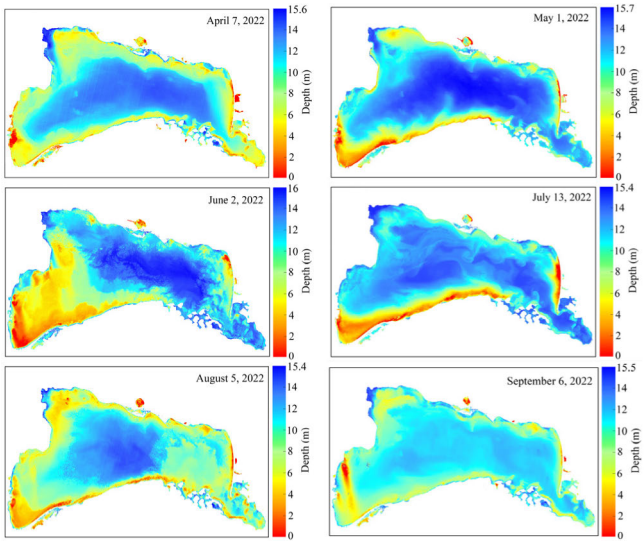


Fig. 14. Bottom depth of the Bosten Lake by using dual-band ratio model.

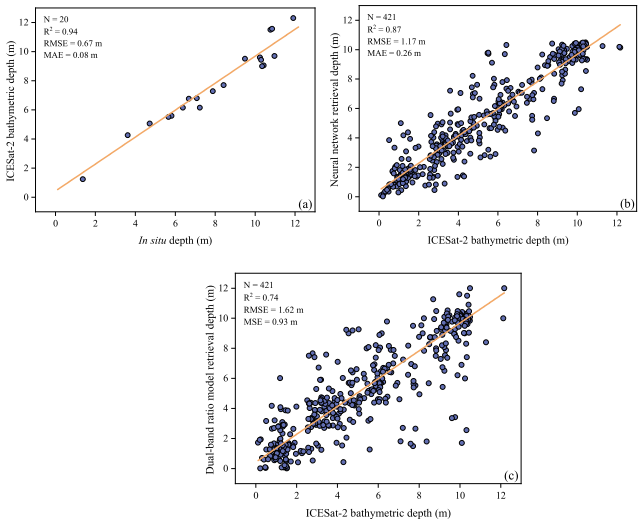


Fig. 15. Error scatter diagrams. (a) ICESat-2 bathymetric depth and in situ water depth. (b) NN retrieved water depth and ICESat-2 bathymetric depth. (c) DBRM retrieval water depth and ICESat-2 bathymetric depth.

NN model, the inverse distance weighted (IDW) interpolation method was applied to obtain the lake's bathymetric map using our measured data (June 27, 2022, in Fig. 13). This alternative method performed well on the land-water interface shore but generated an overestimation in the lake's shallow southern part.

2) DBRM Retrieval Bottom Depth: The water depth of the Bosten Lake was inverted separately using the DBRM, and the bottom depth results are shown in Fig. 14. The uncertainty of the retrieval results is large due to the influence of the lakeshore boundary and local transparency. The bathymetric retrieval errors from August 5, 2022, and September 6, 2022 are large due to local transparency effects.

D. Comparison Between NN and DBRM

A total of 20 in situ water depth data (Fig. 2) were used to further assess the accuracy of the ICESat-2 photon. The in situ water depth data were used to match the corrected photon data by considering the change in water level at different times. The R^2, RMSE, and MAE values of 0.95, 0.67, and 0.08 m, respectively, indicating that the derivation of bathymetry has been made possible by ICESat-2, as shown in Fig. 15(a). Using ICESAT-2 tracks, 421 inverted bathymetry values were extracted to compare the effectiveness of the NN model and DBRM retrieval of bathymetry. The bathymetric depth of the NN inversion was fit to the ICESat-2 bathymetric depth in Fig. 15(b). The water depth of the NN retrieval with ICESat-2 bathymetric depth has an R^2 of 0.87, RMSE of 1.17 m, and MAE of 0.26 m. The water depth of the DBRM retrieval with ICESat-2 bathymetric depth, as shown in Fig. 15(c), has an R^2 of 0.74, RMSE of 1.62 m, and MAE of 0.93 m. The statistical findings demonstrated that, in comparison to the DBRM method, the NN model could offer a more accurate estimate of the water depth.

E. Variation of Bosten Lake Water Storage

The water surface range extracted by Landsat-8 was used to obtain the water column height of the pixel point according to the water depth data retrieved by the NN model. The water column height of the lake was multiplied by the pixel area to obtain the water column volume of the pixel point. The instantaneous water storage of Bosten Lake was obtained by integrating the water column volume of all the pixel points on the lake's surface.

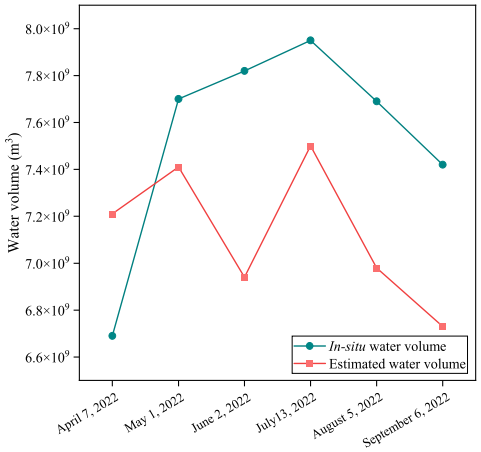


Fig. 16. Variation of water storage in Bosten Lake.

The variation of water storage in the Bosten Lake in different seasons from April to September in 2022 is shown in Fig. 16. The water storage variation was 6.73×10^9 to 7.50×10^9 m^3. However, the in situ water storage variation was 6.69×10^9 to 7.95×10^9 m^3.

V. Discussion

A. Selection of Optical Images

Many multiband imagers can be used to empirically detect water depths [42], [43], [44]. However, only Landsat-8 OLI satellite images were selected for this study. Compared to other multiple sensors, such as MODIS, Sentinel-2, and WorldView-3, Landsat-8 OLI has been widely used in global watercolor remote sensing research [45], [46]. One of the main reasons is that the 30 m spatial resolution of Landsat- 8

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OLI data could provide a detailed characterization of lake topography, which may have highly heterogeneous bottom depths [33]. Moreover, the seven spectral bands of Landsat-8 OLI cover a wide range of bands [47], containing spectral information on depth, water optical properties, bottom substrate (visible domain), and atmospheric aerosols. Since the TOA emission from the Landsat-8 is well calibrated [45], it is sufficient to calculate the Rayleigh-corrected reflectance (ρ_{rc}) using the SeaDAS package.

B. Detection Capability of DBSCAN

The adaptive DBSCAN algorithm was used to discard the outliers prior to water depth correction in processing ATL03 raw data photons [48]. However, the detected underwater signal photons in Fig. 10(a) and (c) show some discontinuous regions with no or only a small number of photons, resulting in discontinuous surface and underwater signals. The few photons in the discontinuous zone have a lower density than the retrieval signal photons, and the detection and filtering process of the adaptive DBSCAN is considered low-reliability photons. As the water depth increases, water column photons density decreases, leading to DBSCAN changes, an adaptive method for the automatic identification and detection of valid signal photons from noisy photons. The discontinuous zone of the signal photons shown in Fig. 10(c) is also caused in this way. In addition, the ICESat-2 L2 ATL03 acquisition time affects the accuracy of the original photon signal [24].

C. Accuracy Estimation of ICESat-2 Bathymetry

Unfortunately, ICESat-2 laser tracks are less available in the deeper water in the middle of Bosten Lake, and ICESat-2 laser tracks are significantly noisy when the water is deeper [Fig. 3(c)]. Therefore, the in situ water depth data were used for lake-bottom retrieval. The photons near the measured data were used to represent the true depth of the lake bottom. In addition, this study used an indirect data correction method to assess the accuracy of ICESat-2. Literature [39] concluded that in calm waters, where the water-body surface is considered flat, the ICESat-2 water depth RMSE was estimated to be 0.43–0.60 m after correcting for water-body refraction effects. The main horizontal error in the laser pulse entering the water column is caused by the refraction effect on the inclined water surface [49]. Therefore, this study further corrects the refraction effect on the water surface and verifies it with the in situ water depth data. In the water depth correction, this study did not consider the impact of tides, a natural phenomenon in coastal areas, because of the celestial action inducing periodic up-and-down sea-level movement [50].

D. NN Model and DBRM Error Analysis

The NN in this study can obtain highly accurate water depths from Rayleigh-corrected TOA reflectance (ρ_{rc}) (430–2300 nm). However, the errors in the NN model fit were related to Landsat-8 image correction and ICESat-2 laser point data on the lake bottom. For Landsat-8 image, atmospheric correction is extremely important for the reliability of water depth estimates [46]. To avoid the influence of aerosols on atmospheric correction, the Rayleigh-corrected TOA reflectance is used as the model input, which avoids the challenges of obtaining high accuracy R_{rs} [23]. In addition, the contribution of Rayleigh scattering can be accurately calculated and eliminated [50]. Since Rayleigh's contribution is an important part of the top-of-atmospheric radiation in the blue-green region, it is also the spectral window where solar radiation can reach the bottom of most clean waters. The Rayleigh-corrected TOA reflectance significantly enhances the water column signal compared to the ρ_t at the top of the atmosphere, which still contains the Rayleigh scattering contribution [51].

The algorithm coefficients (m_1 and m_0) of the DBRM are highly dependent on different geographical and climate characteristics. Therefore, for different regions, a new set of algorithm coefficients is required for a reliable water depth derivation [38]. Compared to the range of bands used by the NN, the DBRM uses only two bands as inputs, which provides lesser coverage in the water depth data. Furthermore, the DBRM does have a greater amount of noise, where the error variability increases with water depth. Moreover, in optically deep water, the bottom reflectance cannot be transferred back to the water surface, which is beyond the detectable range [23]. Therefore, the NN model is more commonly used for extracting water depth information via remote sensing.

Numerous studies have shown that satellite-derived bathymetric accuracy is affected by water column conditions, water clarity, and bottom type at the time of image acquisition [44], [52]. Therefore, water column conditions and atmospheric correction are proven to be the main factors in bathymetric inversion [53]. For the ICESat-2 dataset, the water depth data are still affected by scattering effects in the water column although most of the errors were corrected [54]. In addition, there is a difference of ± 7 days between the acquisition dates of Landsat-8 images and ICESat-2, and the validation (matching image to in situ bathymetry points) can introduce errors due to resolution and geographic location, especially when the lake bottom has a steep slope or a rough.

E. Limitations and Prospects

This study retrieved the photons at the bottom of the lake using ICESat-2 data combined with the adaptive DBSCAN. However, ICESat-2 is noisy when the Bosten Lake water depth exceeds 10 m. We used the measured water depth for lake-bottom depth tracking and then used the photon number near the measured value as the model input. In addition, there were fewer matching points between the real measurement points and ICESat-2 tracks, and the method of tracking data for real measurements had some impact on the accuracy of NN model training. Furthermore, ICESat-2 accuracy can be affected by the local water clarity, bottom type, and water column conditions. As Bosten Lake is a Class II water body [48], the background noise could modify the returned photons. Therefore, in the future, there is a need to develop higher precision algorithms to retrieve the photons returned at the bottom.

The high-precision and high-resolution ICESat-2 data can observe inland waters with a global lateral span greater than

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3 km, and more intensive observations are produced with the increasing latitudes. Therefore, they are especially useful in monitoring inland waters at mid-and high latitudes. ICESat-2 can provide many accurately tracked bathymetry points and water levels for inland waters (such as rivers, lakes, and reservoirs). With the development of high-resolution multispectral imagery, which is especially important for bathymetry points and water level monitoring in remote areas with no or scarce a priori hydrological data [33]. It is expected to have more high-resolution multispectral images, water temperature data, Lidar images at different latitudes, and better bathymetric algorithms in the future. Such developments can improve early warning and response to extreme natural disasters such as floods and droughts. Overall, the effective and fast ICESat-2 filtering method deserves further study, and the machine learning method is worthy of more investigation for bathymetry retrieval.

While this study focused on a rapid calculation of the water storage of the Bosten Lake utilizing data from Landsat-8 and ICESat-2 together with a NN inversion, future studies could assess the causes of changes in lake water storage and their possible impacts over a long-term period. This is particularly important in the Bosten Lake basin, where water resources have a significant impact on the local economy and ecosystem.

VI. Conclusion

This study combined Landsat-8 OLI data and ICESat-2 data and used NN model to retrieve the Bosten Lake water depth, and applied Rayleigh-corrected TOA reflectance (ρ_{rc}) as the model input. This method has proved to be feasible to estimate the water depth of shallow inland water bodies with high accuracy. The algorithm provides a practicable technique for rapid monitoring of the water depth of shallow water bodies in ungauged areas. The R^2 values of the in situ water depth and the corrected ICESat-2 photon data were above 0.95, which indicates a good predictive capability. Better statistical findings have demonstrated that NN inversion is a more useful way of estimating the water depth of Bosten Lake than the conventional dual-band ratio model. The Bosten Lake water storage variation from April to September in 2022 ranged from 6.73×10^9 to 7.50×10^9 m^3. Our results demonstrated the potential of ICESat-2 for inland shallow-water lake bathymetry monitoring. The findings can inform water resources management in inland shallow water bodies and provide a conceptual basis for policymakers to formulate management and conservation policies.

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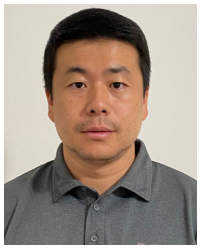
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